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Professor Hamid R. Arabnia
Editor-in-Chief
The Journal of Supercomputing

Re: Manuscript ID feb25704-187e-4f95-8640-5e8c1ca26a94
“Training-Free Ranking from Pairwise Comparisons via Acyclic Graph Construction”

Dear Professor Arabnia,

Thank you for the opportunity to revise my manuscript and for the careful reviews for *The Journal of Supercomputing*. I am submitting the revised manuscript for reconsideration. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I take full responsibility for the scientific content of this revision.

The manuscript has undergone a substantial major revision in response to all four reviewers. The accompanying point-by-point response addresses every comment. The principal changes are:

1. Title and positioning were revised to remove an unqualified scalability claim and to distinguish prior MWFAS components from the contributions of this work.
2. The fixed-topological reinsertion proxy was replaced in the canonical pipeline by exact reachability-aware reinsertion.
3. Principal Tables 4–6 now evaluate the canonical reachability-aware method, OURS-REACH.
4. Inconsistent baseline tables identified by Reviewer 3 were rebuilt from a single canonical data source.
5. Direct classical runtime comparisons were added; the method is slower than most lightweight classical estimators while retaining a substantial end-to-end advantage over the evaluated trained GNN methods.
6. Structural ablations, sensitivity, timeout/coverage, family-aware, and statistical analyses were added.
7. Theory was strengthened (ranking–MWFAS equivalence, reachability correctness, Demetrescu–Finocchi guarantee boundaries, fallback impossibility, and a secondary weighted min-cut repair).
8. The manuscript was substantially rewritten to reduce repetition and improve readability.

I am grateful to you and the reviewers; their comments led to substantial improvements. Detailed responses appear in the accompanying response-to-reviewers document.

Sincerely,

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